

Trust Aware E-Commerce Recommendation Using Brand Risk Analysis

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Abstract—E-commerce platforms generate a continuous stream of user activity through actions such as viewing products, adding to the cart, and purchasing items. Such user activities can provide useful information of the buying intention of the customer during the shopping session. However, using purchase intention for recommending products risks boosting products associated with brands with abnormal behavioral patterns. This paper proposes a trust-aware recommendation method that integrates session-level purchase prediction and brand-level behavioral suspiciousness information.

The purchase prediction module uses aggregated behavioral, price, and time-related features in combination with the user's non-purchase activity during the session. LightGBM, XGBoost, and Random Forest, as base classifiers, and Logistic Regression as the meta-classifier are employed for stacking-based continuous purchase probability estimation. Meanwhile, the brand module uses seven standardized behavioral features and Isolation Forest for identifying unusual patterns to extract suspiciousness and trust information. Both sets of information are integrated into the recommendation module, in which candidate products are gathered using co-view, co-category, and popularity-based retrieval methods.

The retrieved products are re-ranked using candidate relevance, purchase probability, brand trust, and risk penalty. In the experiments, all the candidates have valid Model 1 purchase probabilities while 88.99% have model-derived trust values from Model 2. Given the 1,000 test sessions, the system recommends 29,571 candidate records. The trust-aware ranking increases NDCG@10 from 0.5349 for the purchase-only approach to 0.6078 and results in 0.00% High-Risk Exposure@10. It suggests that behavioral trust information can be leveraged in the recommendation ranking while keeping anomaly detection decoupled from the final fraud decision.

Index Terms—E-commerce recommendation, clickstream analysis, purchase prediction, LightGBM, Isolation Forest, behavioral suspiciousness, trust-aware ranking.

1. INTRODUCTION

E-commerce platforms produce large amounts of data from user interactions during online shopping. During a single shopping session, a customer might browse through several products, switching between different categories, add items to their cart, and abandon the purchase or complete the purchase. All actions that occur during a shopping session are captured as clickstream data. As the events are part of the same shopping session, they can be used to estimate the customer's current purchase intention even in the absence of detailed user information. Prior work has shown that session activity, recent actions, and temporal characteristics can provide valuable information for predicting purchase intention based on clickstream data [1]–[5].

Recommendation systems use customer interaction data to identify products that might be relevant for the current user. Various approaches employ product popularity, item co-occurrence, recent interaction, and session-based models to recommend similar items [6]–[8]. However, such strategies focus predominantly on product relevance and user preferences. Relevance alone does not provide information about the behavioral trust associated with the brand of a recommended product.

This work combines purchase intention and behavioral trust in a unified recommendation pipeline. The first component estimates the purchase probability of the current shopping session. The second component analyzes brand-level behavioral patterns and identifies behavior that is unusual compared with other brands. The output of the second model is represented as behavioral suspiciousness and converted into trust-related information for use during recommendation ranking. This signal is not interpreted as evidence that a brand has committed fraud.

A. Problem Statement

A recommendation system based exclusively on purchase probability, may rank a product as important due to its relevance to the current session, without taking into account the behavioral characteristics associated with its brand. Meanwhile, an independent anomaly detection system may detect anomalous behavior from brands without influencing the products being presented to the customer. Thus, the problem addressed in this work is to develop a recommendation pipeline that jointly considers the purchase intention of the current shopping session and the behavioral trust associated with the brands of candidate products.

B. Limitations of Existing Approaches

Existing session-based recommendation methods focus on retrieving products matching the user's immediate interests and predict relevant items based on current interactions [6]–[8]. Similarly, purchase-intention prediction methods focus on determining if a shopping session is likely to result in a purchase [1]–[4]. Separate research has investigated strange or possibly fraudulent behavior using anomaly detection and graph-based methods [11], [12]. These approaches address different aspects of e-commerce behavior in isolation. A recommender system may therefore provide relevant products without considering available behavioral trust information,

while an anomaly detection model detects unusual behavior without influencing the recommendation list.

C. Proposed Solution

The proposed system combines purchase prediction, brand behavioral analysis and recommendation ranking into a single pipeline. Firstly, the raw clickstream data is cleaned and organized into user sessions. Features describing overall session behavior and recent user actions are then generated for the purchase prediction model. LightGBM, XGBoost and Random Forest are used as base models in a stacking ensemble with Logistic Regression used as the meta-model to combine their outputs. This results in a continuous value representing the purchase probability of the current session.

In parallel, brand-level behavioral features are standardized and processed using Isolation Forest. The model identifies brands with unusual combinations of behavioral characteristics and generates suspiciousness and trust information. The recommendation module retrieves candidate products through session co-view, related-category and global popularity-based channels. The retrieved candidates are then combined with the purchase probability and brand trust information for final ranking.

The final ranking score takes into account candidate relevance, purchase probability, brand trust and a risk-level penalty. Rather than completely removing products associated with higher-risk behavioral categories, the proposed system applies a soft penalty which lowers their ranking position while allowing them to still appear in the candidate pool.

D. Main Contributions

The main contributions of the implemented system are as follows:

- 1) A session-based purchase prediction pipeline that combines aggregate session features and recent user actions using a stacking ensemble.
- 2) A brand-level behavioral suspiciousness model using standardized behavioral features and Isolation Forest.
- 3) Integration of purchase probability and brand trust information directly into the recommendation ranking process.
- 4) Evaluation of the proposed ranking approach against popularity-based, purchase-only, and hard-filter recommendation approaches.
- 5) Validation of model integration, ranking changes, fallback handling, reproducibility, and data-leakage controls.

2. LITERATURE REVIEW

Recommendation systems have been widely employed in e-commerce to assist identifying relevant products from extensive product catalogs. Earlier work of Schafer, Konstan, and Riedl has discussed the application of recommendation techniques in the context of online shopping, as well as the utilization of such data to support the selection of products [6]. With the emergence of the e-commerce platforms, much

of the current literature has been concerned with session-based recommendation, where interactions within the current shopping session can be leveraged to represent the current interests of the customer.

The context of session-based recommendation is particularly useful when information about the customer's global profile is limited. Bauer and Jannach have investigated hybrid session-aware recommendation, based on session interactions plus additional item- and contextual features [14]. Salampasis *et al.* have studied different approaches to session-based recommendation in the context of e-commerce, including co-occurrence-based as well as more sophisticated recommendation techniques [7]. Kersbergen *et al.* have investigated session-based recommendation at scale, and emphasized the importance of low-latency recommendation generation in the context of e-commerce [8]. These works thus show that interactions within the current shopping session can be beneficial in capturing the customer's short-term preferences.

The task of purchase-intention prediction is closely related to the idea of session-based recommendation, yet its purpose is different. Whereas session-based recommendation approaches explicitly choose the next product to recommend, the purpose of purchase-intention prediction is to estimate the likelihood that the current session will conclude with a purchase intent. Esmeli *et al.* have investigated early purchase-intention prediction using online retail session data, and discussed the utilization of behavioral information before the end of the session [3]. Gomes *et al.* have presented customer representations generated from online shopping sessions for purchase-intention prediction purposes [2]. Tokuç and Dag have compared different clickstream representations and machine learning models for the task of purchase prediction [1]. Their approach has considered aggregate session information, recent actions, and hybrid representations; these are relevant to the feature representation adopted in the proposed system.

Tree-based machine learning methods are widely applied in the context of structured prediction tasks. LightGBM, as introduced by Ke *et al.*, is a gradient boosting decision tree framework designed for efficient model training and prediction [9]. In the implemented system, LightGBM is combined with XGBoost and Random Forest, as opposed to being applied as a single predictor. The probability outputs from these base learners are provided to a Logistic Regression meta-model, as part of the stacking procedure. In this way, the final predictor can leverage the information obtained from multiple base learners.

An area of research closely related to purchase-intention prediction is that of anomaly detection. Isolation Forest, proposed by Liu, Ting, and Zhou, is able to identify unusual observations from the given set by isolating them from the remainder of the observations [10]. Much of the literature on e-commerce has investigated both behavioral and graph-based approaches to detect unusual and potentially fraudulent behavior. For example, eFraudCom has applied competitive graph neural networks to detect fraud in e-commerce, whereas behavioral graph-based approaches have exploited interaction

relationships to detect such behavior [11], [12]. On the other hand, the objective of Model 2 in the present work is different from these fraud-detection systems; it uses Isolation Forest to detect unusual brand-level behavioral patterns, and treats the resulting output as the behavioral suspiciousness/trust signal.

Trust-related information can be considered to be relevant when designing recommendation systems. Cabrera-Sánchez *et al.* have investigated factors affecting the use of online recommendation systems in the context of e-commerce [13]. Some research on the topic of session-aware recommendation has also assessed the influence of popularity and feedback properties on offline recommendation evaluation [14]. These works thus provide motivation for considering information beyond product relevance when recommending products.

Based on the reviewed literature, the phenomenon of session behavior has been extensively analyzed both for recommendation and purchase-intention prediction. However, the context of anomaly detection has been mainly pursued as an endeavor to detect unusual e-commerce behavior. The implemented system thus connects these topics by utilizing the current-session purchase probability along with the brand-level behavioral trust when re-ranking recommendation candidates. Thus, the anomaly-related output is not perceived as an exclusive detection signal, but rather as one of the factors influencing the order of recommendation candidates.

3.METHODOLOGY

The proposed system is developed as a combination of purchase prediction, brand-level behavioral analysis, and recommendation ranking. The methodology is divided into three major parts. The first part estimates the probability that a user will make a purchase based on the actions performed during a session. The second part studies the behavior associated with different brands and generates suspiciousness and trust values. The third part uses these outputs along with product relevance to generate the final recommendation list.

The processing begins with the raw e-commerce clickstream data. The data is first cleaned and arranged into user sessions. Features are then extracted from the session activity and given to the purchase prediction model. At the same time, brand-level behavioral features are generated for the second model. Product candidates are obtained using different retrieval methods and are combined with the outputs from both models. The final candidates are assigned a trust-aware score and sorted to produce the recommendations.

A. Dataset and Preprocessing

The system uses the `2019-Oct.csv` e-commerce clickstream dataset. Each record contains information such as event time, event type, product ID, category information, brand, price, user ID, and user session ID. These fields are used to understand the sequence of actions performed by users while browsing the platform.

Before model training, the dataset is cleaned to remove duplicate records. Missing brand values are represented as unknown, while the category-code field is not used because

Table 3.1
DATASET AND EVALUATION STATISTICS

Item	Validated Value
Raw clickstream file	2019-Oct.csv
Product catalog	30,223 products
Brand catalog / Model 2 coverage	765 brands
Held-out Model 1 sessions	1,069
Recommendation evaluation sessions	1,000
Recommendation candidates	29,571
Average candidates/session	29.57
Unique products in integrated candidates	4,960
Unique brands in integrated candidates	733

of the missing values present in that field. The event time is converted into datetime format. After this conversion, the records are sorted according to the user session and event time. This ordering is required because the sequence in which the user performs actions is considered during feature generation.

The validated product catalog contains 30,223 products from 765 brands. Model 2 produces trust information for these 765 scored brands. For recommendation testing, 1,000 sessions are used and 29,571 candidate rows are generated. This corresponds to an average of 29.57 candidate products for each evaluation session.

B. Sessionization and Purchase Label

The records are grouped using the `user_session` identifier. Within every session, the events are arranged according to their timestamp. This maintains the actual order of user activity and allows the model to distinguish between different sequences of actions.

Sessions with very few interactions are excluded since they do not contain enough information for reliable behavioral analysis. A session is labelled as a purchase session when at least one purchase event is present. Sessions without a purchase event are assigned the non-purchase label.

After the target label is generated, the purchase event is removed from the model input. This step is important because keeping the purchase event would directly reveal the outcome that the model is expected to predict.

C. Session Feature Engineering

The system does not treat every click as a separate training sample. Instead, the activities performed during a complete session are summarized into a set of features. This provides the model with information about the overall behavior of the user during that session.

The session features include the total number of events, number of product views, number of cart actions, number of unique products, number of unique categories, and number of unique brands. Price information is represented using statistics such as the average, minimum, and maximum product price observed in the session.

The time-related features include session duration, starting hour, day of the week, and a weekend indicator. The recent activity of the user is also included because the latest actions can provide useful information about purchase intent. The

five most recent non-purchase actions are stored as sequential features. A product view is represented by 1 and a cart action by 2. When fewer than five actions are available, the unused positions are filled with 0.

This combination of aggregate and recent-action features allows the model to consider both the general behavior of the session and the user’s latest interactions.

D. User Intent Prediction Using LightGBM and Stacking

Four classifiers are implemented for predicting purchase intent: Logistic Regression, Random Forest, XGBoost, and LightGBM. LightGBM is configured with 200 estimators, a learning rate of 0.05, 31 leaves, and balanced class weighting. XGBoost uses 200 estimators, a learning rate of 0.05, and a maximum depth of 5. Random Forest is configured with 300 trees, a maximum depth of 12, and balanced class weighting.

The final purchase prediction is obtained using a stacking approach. LightGBM, XGBoost, and Random Forest are used as the base models. Their predictions are then given to Logistic Regression, which acts as the meta-model. Five-fold stratified cross-validation is used during training to generate out-of-fold probability values for the base models.

For a new session, each fold-specific base model produces a prediction. These predictions are averaged and passed to the Logistic Regression meta-model. The final result is a continuous purchase probability rather than only a binary purchase or non-purchase output.

The purchase probability for a session s is represented as $P(\text{purchase} \mid s)$ and is later used during recommendation ranking.

$$P(\text{purchase} \mid s) = \text{Stacking}(\text{LightGBM}(s), \text{XGBoost}(s), \text{RandomForest}(s)). \quad (1)$$

E. Threshold Optimization

The probability generated by the stacking model is continuous, so a threshold is required only when the purchase prediction is evaluated as a binary classification problem. The recommendation stage continues to use the original probability value.

A threshold of 0.5 was considered initially. However, different threshold values were tested because the dataset contains an imbalance between purchase and non-purchase sessions. Among the tested values, a threshold of 0.90 produced the highest F1-score.

At the selected threshold of 0.90, the purchase prediction model achieved an accuracy of 0.9371, precision of 0.6364, recall of 0.3500, and F1-score of 0.4516. ROC-AUC and PR-AUC are calculated directly from the probability outputs and therefore are not affected by the selected classification threshold.

F. Brand Behavioral Feature Engineering

The second model works with brand-level information rather than individual user sessions. The clickstream activity is

Table 3.2
BEHAVIORAL SUSPICIOUSNESS AND TRUST CATEGORIES

Risk Level	Brands	Percentage	Trust Range
Low	710	92.81%	0.7000–0.9995
Medium	49	6.41%	0.4000–0.6999
High	6	0.78%	0.1500–0.3999
Total	765	100%	0.1500–0.9995

aggregated for each brand to describe its observed behavioral pattern. Seven behavioral features are used for this purpose.

Before applying the anomaly detection model, the generated features are standardized using a fitted StandardScaler. Standardization puts the different feature values on a comparable scale and provides the processed input required by the anomaly detection stage.

The available validation information confirms that seven features are used, but it does not provide their individual feature names. Therefore, no additional feature names are specified here beyond the information available from the implementation.

G. Behavioral Suspiciousness Estimation Using Isolation Forest

Isolation Forest is used in Model 2 to identify brands that show unusual combinations of behavioral features. The implemented Isolation Forest contains 100 estimators and uses a contamination value of 0.08.

The output of this model is used to obtain suspiciousness and trust information for the recommendation stage. The purpose of this step is to identify unusual behavioral patterns rather than to directly declare a brand as fraudulent. Since the clickstream dataset does not contain verified fraud labels for all brands, the resulting suspiciousness value is treated as an indicator of unusual behavior.

The validated Model 2 output contains 765 scored brands. The suspiciousness values range from 0.0005 to 0.8500, while the trust values range from 0.1500 to 0.9995. Based on the stored trust and risk information, the brands are divided into Low, Medium, and High risk categories.

The exact transformation used in the implementation to convert the raw Isolation Forest output into the stored suspiciousness and trust values is not specified in the available validation material. Therefore, no additional transformation equation is introduced.

H. Candidate Product Retrieval

Once the purchase probability and brand-level information are available, the recommendation module creates a set of possible products for each session. Three retrieval methods are used to form this candidate set.

The first method retrieves products that have been co-viewed with products present in the current session. The second method retrieves products from related categories. The third method uses globally popular products so that the candidate set also contains products that may be relevant even when limited session information is available.

Each retrieval method produces a relevance value for the products it returns. If a product appears through more than one retrieval method, duplicate entries are removed using the session and product identifiers. The highest available relevance value is retained. The final source relevance value is represented by $R(s, p)$ and is restricted to the range $[0.1, 1.0]$.

I. Trust Integration and Fallback Handling

The outputs from the two models are combined with the candidate product data. Model 1 is connected to the candidate data using the `user_session` identifier. Model 2 is connected using the brand associated with each product.

The integrated data contains the purchase probability, trust score, and suspiciousness values in the range $[0, 1]$. During the validated evaluation, all 29,565 evaluated candidates received an actual purchase probability from Model 1.

Model 2 supplied real trust values for 26,309 candidates, which represents 88.99% of the evaluated candidates. For the remaining candidates, fallback values are used when sufficient brand-level behavioral information is not available.

Among the evaluated candidates, 2,460 candidates, or 8.32%, had unknown brand information. These candidates were assigned a trust value of 0.50. A further 796 candidates, or 2.69%, belonged to brands having fewer than five lifetime events and were assigned a trust value of 0.60.

These fallback values represent limited behavioral evidence. They are not used as evidence that the corresponding brands are fraudulent.

J. Trust-Aware Recommendation and Re-ranking

The final ranking combines product relevance with the predicted purchase probability and the trust information obtained from Model 2. A risk penalty is also included so that the behavioral information can affect the position of a product in the recommendation list.

The final score for a product p in session s is calculated as follows:

$$S(s, p) = R(s, p) [1 + \alpha P(\text{purchase} \mid s)] \times \text{Trust}(b)^\beta \Psi(b). \quad (2)$$

Here, $R(s, p)$ is the relevance value of product p for session s . The term $P(\text{purchase} \mid s)$ is the purchase probability obtained from Model 1. $\text{Trust}(b)$ represents the trust value of the brand b , and $\Psi(b)$ represents the penalty associated with its risk category. The implementation uses $\alpha = 1.0$ and $\beta = 1.0$.

The risk penalty is set to 1.00 for Low-risk brands, 0.75 for Medium-risk brands, and 0.20 for High-risk brands. Therefore, a product with high relevance and high purchase probability can receive a higher ranking, while lower brand trust or a higher risk category can reduce its final score.

The system uses a soft penalty rather than removing products completely. Products belonging to the High-risk category receive a stronger reduction in their score, whereas Medium-risk products receive a smaller reduction. This allows the recommendation process to retain candidate products while still considering the behavioral information generated by Model 2.

K. Overall Algorithm

The complete workflow of the proposed system can be summarized in the following steps:

- 1) Load the e-commerce clickstream dataset and perform the required preprocessing.
- 2) Group the events using the `user_session` identifier and arrange the events in chronological order.
- 3) Generate session-level features, including aggregate activity, price-related information, time information, and recent user actions.
- 4) Use the trained stacking model to obtain the purchase probability for each session.
- 5) Aggregate the available behavioral information for each brand and generate the corresponding suspiciousness and trust values using Isolation Forest.
- 6) Retrieve possible recommendation candidates using co-view, co-category, and global popularity information.
- 7) Join the candidate products with the purchase probability from Model 1 and the brand-level trust information from Model 2.
- 8) Apply the defined fallback values when reliable brand-level behavioral information is not available.
- 9) Calculate the trust-aware recommendation score for every candidate product.
- 10) Sort the candidate products according to their calculated scores.
- 11) Select the Top- K products from the ranked candidate list.
- 12) Evaluate the recommendations using relevance, trust, exposure, sensitivity, and ranking-based measures.

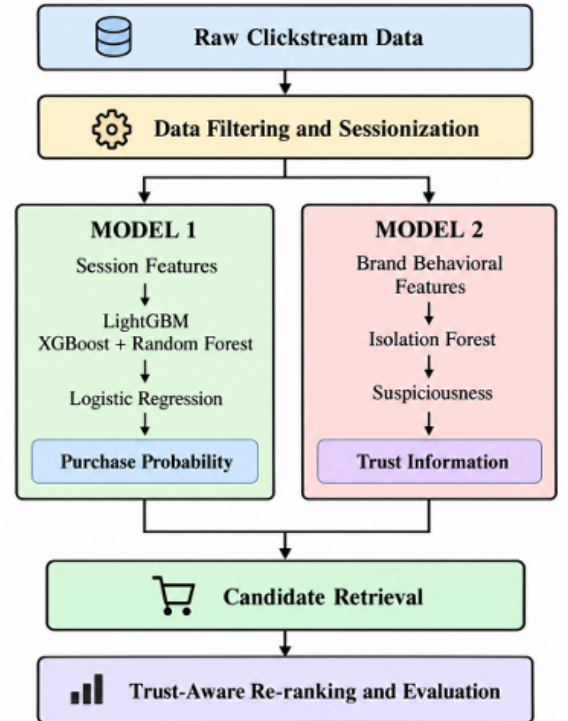


Figure 3.1. Overall System Architecture

4.RESULTS

The proposed system was evaluated to benchmark the performance of the purchase prediction model, the behavioral suspiciousness and trust model, and the final trust-aware recommendation pipeline. The evaluation also verifies that the outputs of both models are indeed being incorporated into the recommendation ranking process.

A. Dataset and Evaluation Statistics

The recommendation system evaluated on 1,000 sessions, which produced 29,571 candidate rows during the candidate generation stage. Both the purchase prediction and behavioral suspiciousness models were applied and their scores used to derive the final recommendation rankings.

Table 4.1
DATASET AND EVALUATION STATISTICS

Metric	Value
Evaluation Sessions	1,000
Candidate Rows	29,571

B. Purchase Prediction Results

Four implemented models were evaluated for purchase prediction: LightGBM, XGBoost, Random Forest, and the Stacking Ensemble. Their performance was measured using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and log loss. The results are given in Table 4.2.

The results show that the models differ significantly in the results. Random Forest recorded the highest accuracy, precision, and F1-score among individual models, while XGBoost recorded the highest recall. Finally, the Stacking Ensemble recorded the highest ROC-AUC and PR-AUC in the comparison, although its accuracy and precision was lower than with Random Forest.

For the final binary classification, an optimized decision threshold is considered, rather than the default threshold. The selected threshold along with its precision, recall, and F1-score is presented in Table 4.3.

At the selected threshold of 0.90 the optimized classifier achieved an accuracy of 0.9371, precision of 0.6364, recall of 0.3500, and F1-score of 0.4516. Increasing the threshold makes the classifier more selective, and thus improves precision but decreases recall.

For the recommendation generation process, the continuous stacking probability is maintained, rather than converting it to a binary decision, so that the estimated purchase intent contributes directly to the ranking score.

Table 4.2
PURCHASE PREDICTION MODEL RESULTS

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Log Loss
LightGBM	0.8483	0.2470	0.5125	0.3333	0.7812	0.3695	0.3580
XGBoost	0.8316	0.2316	0.5500	0.3259	0.7791	0.3494	0.4108
Random Forest	0.8945	0.3300	0.4125	0.3667	0.7814	0.3664	0.3342
Stacking Ensemble	0.7946	0.1966	0.5750	0.2930	0.7845	0.3861	0.5244

Table 4.3
OPTIMIZED PURCHASE CLASSIFICATION THRESHOLD

Threshold	Precision	Recall	F1-score
0.90	0.6364	0.3500	0.4516

C. Behavioral Suspiciousness and Trust Results

The second model operates at the level of brands, providing behavioral suspiciousness information, which is further converted into a trust value that informs the recommendation component. The resulting brands are divided into low-, medium- and high-risk behavioral categories.

Six brands in the high-risk category were identified by evaluation, as shown in Table 4.4.

Table 4.4
BEHAVIORAL RISK DISTRIBUTION

Risk Category	Number of Brands
High Risk	6

These categories represent the model’s generated behavioral classification. Thus, a high-risk category reflects unusual behavioral characteristics identified by the implemented model, and should not be interpreted as a legal conclusion or proof of fraudulent activity.

D. Evidence That Both Models Affect the Recommendation

An integration check was performed to verify that the actual outputs of both models were used during the recommendation generation process. This served to also determine whether predefined fallback values were having a major influence on the final ranking.

The integration results are presented in Table 4.5.

Table 4.5
MODEL INTEGRATION AND RANKING INFLUENCE

Integration Measure	Result
Model 1 real probability usage	29,565/29,565 (100.00%)
Model 1 fallback intent usage	0.00%
Model 2 real trust usage	26,309/29,565 (88.99%)
Model 2 fallback trust usage	3,256/29,565 (11.01%)
Unknown-brand fallback	2,460 candidates (8.32%)
Low-confidence-brand fallback	796 candidates (2.69%)
Candidate rank positions displaced by trust	2,972
High-trust items promoted into top 3	628 instances
Medium/high-risk items demoted	1,623 instances

All 29,565 evaluated candidates received an actual purchase probability from Model 1, and therefore no fallback intent value was required. Model 2 supplied actual trust values for 26,309 candidates or 88.99% of the evaluated candidates.

For the remaining candidates, fallback values were required since certain brands were unknown or did not have sufficient evidence. The trust component changed 2,972 candidate ranking positions. Additionally, 628 high-trust items were promoted into the top three positions, while 1,623 medium or high-risk items were demoted. These results show that the trust information has a direct influence on final ranking.

E. Recommendation Baseline Comparison

The trust-aware ranking method was compared against popularity-based ranking, purchase-only ranking, and hard-filter ranking. The comparison considered Top-5 and Top-10 recommendations using HitRate, Precision, Recall, NDCG, Average Trust, and High-Risk Exposure. The complete results are presented in Table 4.6.

For both $K = 5$ and $K = 10$, the trust-aware approach recorded the highest measured values for HitRate, Precision, Recall, and NDCG among the evaluated ranking strategies. At $K = 10$, NDCG increased from 0.5349 with purchase-only ranking to 0.6078 with trust-aware ranking.

Additionally, Average Trust@10 increased from 0.5999 to 0.6654. The hard-filter approach recorded zero high-risk exposure, yet its HitRate@10 and NDCG@10 values were lower than those obtained using the trust-aware method.

The comparison shows the difference between completely removing candidates and reducing their ranking priority. The implemented trust-aware approach applies a soft penalty, allowing relevance and purchase intent to remain part of the ranking, while behavioral trust influences the final position.

Table 4.6
RECOMMENDATION BASELINE COMPARISON

Metric	Popularity	Purchase-only	Hard-filter	Trust-aware
HitRate@5	0.1400	0.7630	0.7520	0.8220
Precision@5	0.0280	0.1554	0.1532	0.1672
Recall@5	0.1400	0.7630	0.7520	0.8220
NDCG@5	0.0896	0.4826	0.4759	0.5685
Average Trust@5	0.5629	0.6158	0.6223	0.6677
High-Risk Exposure@5	1.16%	1.36%	0.00%	0.00%
HitRate@10	0.2050	0.9210	0.9060	0.9410
Precision@10	0.0207	0.0937	0.0922	0.0956
Recall@10	0.2050	0.9210	0.9060	0.9410
NDCG@10	0.1107	0.5349	0.5269	0.6078
Average Trust@10	0.5752	0.5999	0.6062	0.6654
High-Risk Exposure@10	0.98%	1.21%	0.00%	0.00%

F. Fallback-Only Ablation

An additional ablation evaluated a fallback-only configuration against the configuration using the actual machine learning outputs. The results are presented in Table 4.7.

Table 4.7
FALLBACK-ONLY ABLATION

Metric	Fallback-only	Real ML Models	Change
HitRate@10	0.9210	0.9410	+0.0200
NDCG@10	0.5349	0.6078	+0.0729
Average Trust@10	0.6000	0.6999	+0.0999
High-Risk Exposure@10	1.21%	0.00%	-1.21 pp

Using the actual machine learning outputs increased HitRate@10 by 0.0200 and NDCG@10 by 0.0729, while Average Trust@10 increased by 0.0999, and High-Risk Exposure@10 decreased from 1.21% to 0.00%.

The ablation therefore provides additional evidence that the recommendation results are influenced by the actual model predictions rather than being generated predominantly from fixed fallback values.

G. High-Risk Ranking Case Study

A controlled example involving three products was used to illustrate the impact of the trust signal on the final ranking. All three candidates were assigned the same relevance value of 1.0 and purchase probability of 0.80, while their trust values differed according to their behavioral risk categories.

The “globber” brand had a trust value of 0.198 and was categorized as High Risk, “samsung” was assigned a trust value of 0.582 and categorized as Medium Risk, and finally “apple” had a trust value of 0.999 and was categorized as Low Risk.

Without the trust component, all three candidates received the same purchase-only score of 1.8000. Once the trust-aware ranking calculation was applied, their scores became 0.0713, 0.7857, and 1.7982 for globber, samsung, and apple respectively. The resulting ranking scores are illustrated in Fig. G.

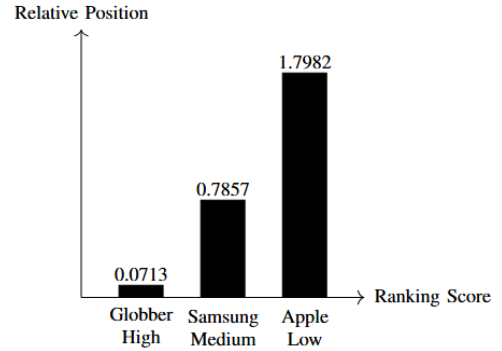


Figure 4.1. Effect of Trust on a Controlled Ranking Example

The example shows that the trust component has the ability to change the relative ordering of candidates, even with identical relevance and purchase probability.

5.FUTURE DISCUSSION / FUTURE WORK

The current implementation offers an end-to-end offline trust-aware recommendation pipeline, but there are certain ways in which it could be improved before deployment in a real e-commerce environment.

First, behavioral modelling could be enhanced by using longer and more detailed interaction histories. Instead of only using aggregated brand behavior, future versions could model how behavior changes over time and how patterns differ across product categories. This could reduce false suspiciousness for brands that naturally have unusual but legitimate activity.

Second, additional data sources could be included. Product reviews, return behavior, seller history, price changes, delivery information and verified transaction outcomes could provide stronger evidence than clickstream behavior alone. These signals could be added with care so that the trust model is not biased towards brands with higher traffic.

Third, the recommendation model could be extended beyond the current multi-channel candidate retrieval and multiplicative re-ranking formula. Learning-to-rank methods, neural session models, graph-based recommendation and hybrid rec-

ommenders could be evaluated while retaining the trust signal as an additional ranking objective.

Fourth, the system could be moved towards real-time recommendation. The current validation processes batches of session data and saved model outputs. A production version could update session features after every interaction and retrieve new candidates with low latency. The brand trust model could also be updated periodically as new behavioral evidence becomes available.

Fifth, trust estimation should be calibrated and validated with stronger ground truth. Human review, verified seller information and confirmed platform outcomes could help determine whether high suspiciousness actually corresponds to meaningful risk. Until such evidence is available, the system should continue to describe the output as behavioral suspiciousness or trust estimation rather than fraud detection.

Finally, explainability should be added to the recommendation interface. A user or platform administrator should be able to see why an item was ranked highly, for example because it matched the current session and had a high purchase probability, while another item was demoted because its associated brand had a lower behavioral trust score. This would enable the system to be more easily inspected and improved.

6. CONCLUSION

The proposed system has been created and validated as an e-commerce recommendation system that leverages clickstream data to combine purchase intent and behavioral trust. The purchase prediction component synthesizes the session behavior into aggregate and recent-action features and applies LightGBM, XGBoost and Random Forest inside a stacking ensemble. The second component uses Isolation Forest on brand-level behavioral features to provide suspiciousness and trust information which are later integrated to trusted candidate retrieval and trust-aware re-ranking.

The experimental results demonstrate that the proposed integration is indeed a practice rather than a theory. All the evaluated recommendation candidates received real Model 1 purchase probabilities, 88.99% got real Model 2 trust values and trust resulted in thousands of candidate rank changes. The proposed ranking has achieved NDCG@10 of 0.6078 compared with 0.5349 for purchase-only ranking and had resulted in 0.00% High-Risk Exposure@10 in the reported baseline comparison. The system therefore can be considered as a practical way to include behavioral trust information in e-commerce recommendation while keeping an adequate distinction between unusual behavior and proven fraud.

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